

TAHA TARKHANI

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Education

University Of Ottawa

Expected Fall 2027

Bachelor of Applied Science in Software Engineering

Ottawa, Ontario

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Object Oriented Programming, Software Engineering, Software Design and Architecture, Algorithm Design and Analysis, Database Systems, Networks

Experience

National Defence Canada

Sep 2025 – Dec 2025 & May 2026 – Aug 2026

Software Engineer Intern

Ottawa, ON

- Built full-stack features for **Space-COP** platform serving military analysts tracking space operations, delivering real-time data with **sub-200ms** response times to end users.
- Accelerated release cycle by **45%** by designing and implementing automated **CI/CD** pipelines, enabling daily deployments while maintaining **99.7%** uptime for critical defense operations.
- Improved code reliability by increasing test coverage from **32%** to **78%** through comprehensive **JUnit** and integration tests, preventing mission-critical bugs before production.
- Partnered with military analysts, infrastructure teams, and security specialists to harden the platform, ensure government compliance standards, and modernize legacy system components.

National Research Council Canada

Jan 2025 – Apr 2025

Data Analyst Intern

Ottawa, ON

- Automated data processing with **Python** scripts for scientific research datasets (**PostgreSQL/MySQL**), reducing manual data entry time by **70%** through **ETL** workflows at scale.
- Designed the data pipeline's technical architecture, implementing **SQL** query optimization, error handling, and validation checks that enabled clean, seamless integration with downstream analytics tools.
- Partnered with research teams to validate data quality and formalize **ETL** process documentation, improving pipeline reliability and easing knowledge transfer for future team members.

Projects

Request Tracer | *Java, Spring Boot, React, Docker, REST APIs*

- Built and verified end-to-end correlation ID propagation across **4** independent **Spring Boot** services (Gateway, Orders, Inventory, Log Aggregator), confirming one identical trace ID across all **6** logged events per request.
- Containerized the full system as **5** Docker images with multi-stage builds and environment-driven service URLs, orchestrated via **Docker Compose** for one-command startup with zero code changes between environments.
- Developed a **React** dashboard (Vite) rendering live traces as a waterfall timeline, with real end-to-end latency measured across **7** Docker runs showing a one-time **1565ms** cold-start versus a stable **64-119ms** steady state.

Orbit Tracker | *Python, Flask, Skyfield, globe.gl, REST APIs*

- Engineered a Python service using **Skyfield** (SGP4 propagation) to track **30** live satellites from **CelesTrak's** public API, validating results against known orbital data (e.g., GOES-16 altitude within **0.02%** of its true geostationary orbit).
- Cached TLE data to cut repeat fetch times from **2.43s** to under **10ms**, with position calculations running in **1.24ms** and full API responses served in **5-7ms**.
- Implemented a pass-prediction **REST API** that found **84** upcoming satellite passes across a 5-day window in **106.6ms**, visualized on an interactive **3D** globe using **globe.gl**.

Cost Monitor | *Python, Flask, AWS Cost Explorer, Boto3, DynamoDB, Chart.js*

- Built an AWS cost monitoring dashboard in **Flask** that ingests billing data via the **Cost Explorer API** and persists daily snapshots in **DynamoDB**, correctly handling the API's real access and indexing failure states.
- Designed and validated an anomaly detection system against test data, correctly distinguishing a **2.9%** fluctuation (unflagged) from an **83.2%** cost spike (flagged) using configurable thresholds.
- Implemented an in-memory caching layer achieving **0.0ms** cache-hit response times, with full dashboard loads measured at **45-186ms** across real local testing.

Technical Skills

Languages: Python, Java, JavaScript, C++, SQL, C#, TypeScript, HTML/CSS

Technologies & Tools: React, Spring Boot, Node.js, Flask, Docker, AWS, PostgreSQL, MySQL, Git, REST APIs, CI/CD, GitHub Actions, JUnit, Jest, Boto3, DynamoDB, Skyfield, Chart.js, globe.gl